

Exploratory Data Analysis (EDA) Report

Dynamic Pricing Strategies for Fitness Classes based on demand, time and location.

Objective

The objective of this Exploratory Data Analysis (EDA) is to analyze booking patterns, pricing trends, customer demand, and location-wise performance of fitness classes. The analysis aims to understand how demand changes across different time slots, locations, and activities. The insights obtained from this EDA will help build an effective dynamic pricing strategy that improves occupancy and maximizes revenue.

Dataset Description

The dataset contains historical booking records of fitness classes conducted at different activity locations. It includes information related to class schedules, customer bookings, pricing, and seating capacity. The dataset is used to analyze customer demand, time-based booking patterns, location-wise performance, and revenue generation. These insights help support the development of an effective dynamic pricing strategy for fitness classes.

Dataset Summary

Item	Value
Total Records	(3289)
Total Features	(12)
Numerical Columns	Price, Number Booked, MaxBookees, Occupancy Rate, Revenue
Categorical Columns	ActivitySiteID, ActivityDescription, Day, Month, TimeSlot
Date & Time Columns	BookingEndDate, BookingStartTime

Data Cleaning Summary

The dataset was cleaned and prepared before performing the exploratory data analysis. The following preprocessing steps were carried out to improve data quality and ensure accurate analysis.

- Converted **BookingEndDate** to the correct date format.
- Converted **BookingStartTime** to the appropriate time format.
- Checked for missing values and handled them where required.
- Checked for duplicate records to avoid repeated information.
- Detected and reviewed outliers in numerical columns.
- Verified and corrected data types for all columns.

Statistical Analysis

The statistical analysis provides a summary of the numerical features in the dataset. It helps understand the overall distribution, central tendency, and variation of booking data, pricing, and class capacity before performing detailed analysis.

```
data.describe().T
```

	count	mean	min	25%	50%	75%	max	std
BookingEndDate	3289	2018-05-16 08:58:57.610215	2018-04-01 00:00:00	2018-04-24 00:00:00	2018-05-16 00:00:00	2018-06-08 00:00:00	2018-06-30 00:00:00	NaN
BookingStartTime	3289	1900-01-01 13:38:29.425358	1900-01-01 06:30:00	1900-01-01 10:00:00	1900-01-01 12:30:00	1900-01-01 17:45:00	1900-01-01 20:15:00	NaN
MaxBookees	3289.0	32.179386	2.0	24.0	30.0	35.0	70.0	15.092156
Number Booked	3289.0	17.088781	1.0	10.0	16.0	24.0	64.0	9.504098
Price (INR)	3289.0	1850.231377	499.0	1299.0	1499.0	2499.0	3999.0	791.20729

The statistical summary shows that the dataset contains a wide range of values for **Price**, **Number Booked**, and **MaxBookees**. The variation across these features indicates differences in customer demand, class capacity, and pricing among fitness classes.

Feature Engineering

The objective of feature engineering is to create new meaningful features from the existing data that help improve business analysis and support the development of an effective dynamic pricing strategy.

Features Created

1. Month

Description:

The month was extracted from the booking date to analyze monthly booking trends.

Business Purpose:

It helps identify seasonal demand and booking patterns.

2. Day

Description:

The day of the week was extracted from the booking date.

Business Purpose:

It helps understand customer booking behavior on different days.

3. Hour**Description:**

The hour was extracted from the booking start time.

Business Purpose:

It helps identify peak booking hours during the day.

4. TimeSlot**Description:**

Booking hours were grouped into Morning, Afternoon, and Evening time slots.

Business Purpose:

It helps compare customer demand across different parts of the day.

5. Occupancy Rate**Description:**

Occupancy Rate measures the percentage of available seats that were booked.

Formula:

$\text{Occupancy Rate} = (\text{Number Booked} / \text{MaxBookees}) \times 100$

Business Purpose:

It helps identify high-demand and low-demand fitness classes.

6. Revenue**Description:**

Revenue was calculated using the class price and the number of bookings.

Formula:

$\text{Revenue} = \text{Price} \times \text{Number Booked}$

Business Purpose:

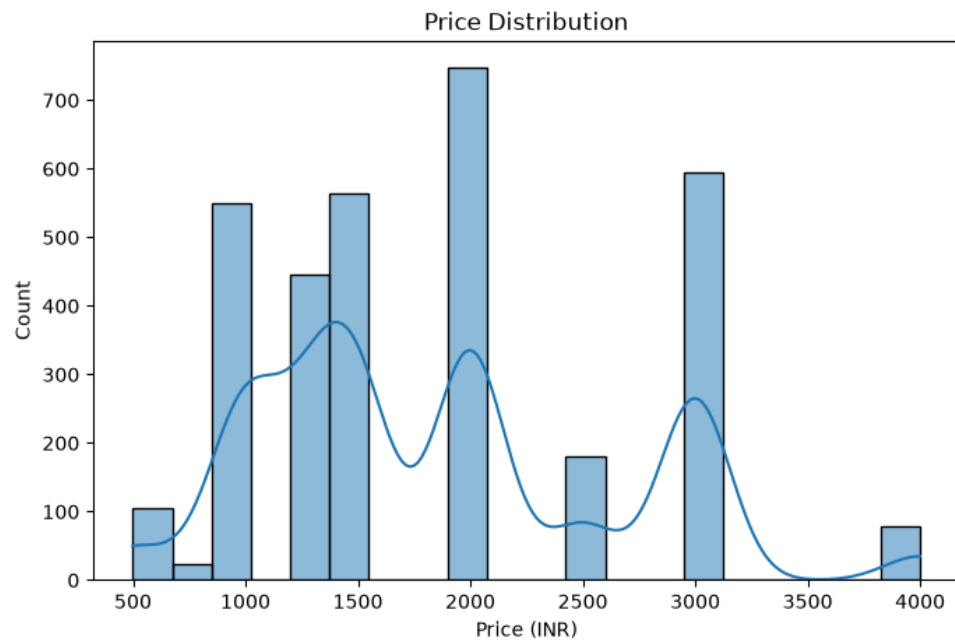
It helps identify the activities, time slots, and locations that generate the highest revenue.

Observation

The newly created features provide additional business insights beyond the original dataset. They help analyze booking trends across time, demand, occupancy, and revenue more effectively.

Univariate Analysis

1. Price Distribution (Histogram)



Objective

To understand the distribution of fitness class prices.

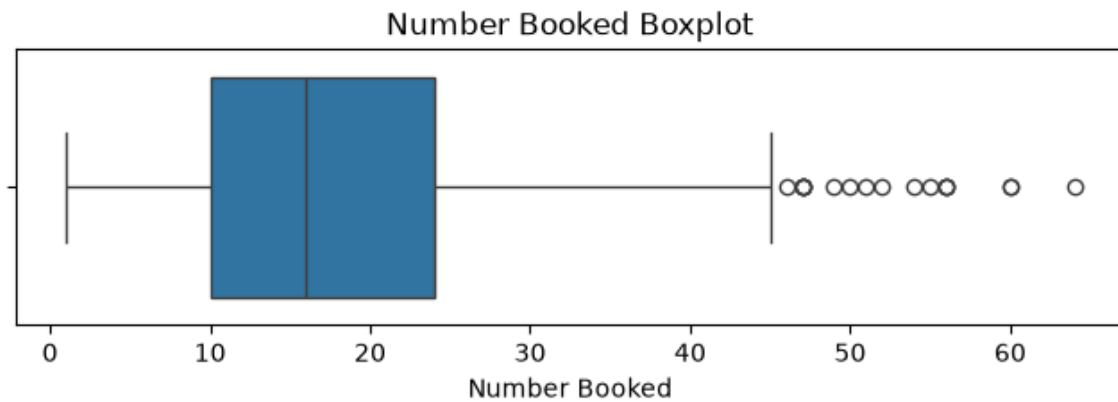
Observation

Most fitness class prices are concentrated between **₹1,000 and ₹3,000**. Only a small number of classes have very low or very high prices.

Business Insight

The organization mainly follows a standard pricing strategy. Premium-priced classes should be monitored regularly to ensure they continue to attract customer demand and generate higher revenue.

2. Number Booked (boxplot)



Objective

To identify unusual booking patterns and extreme booking values.

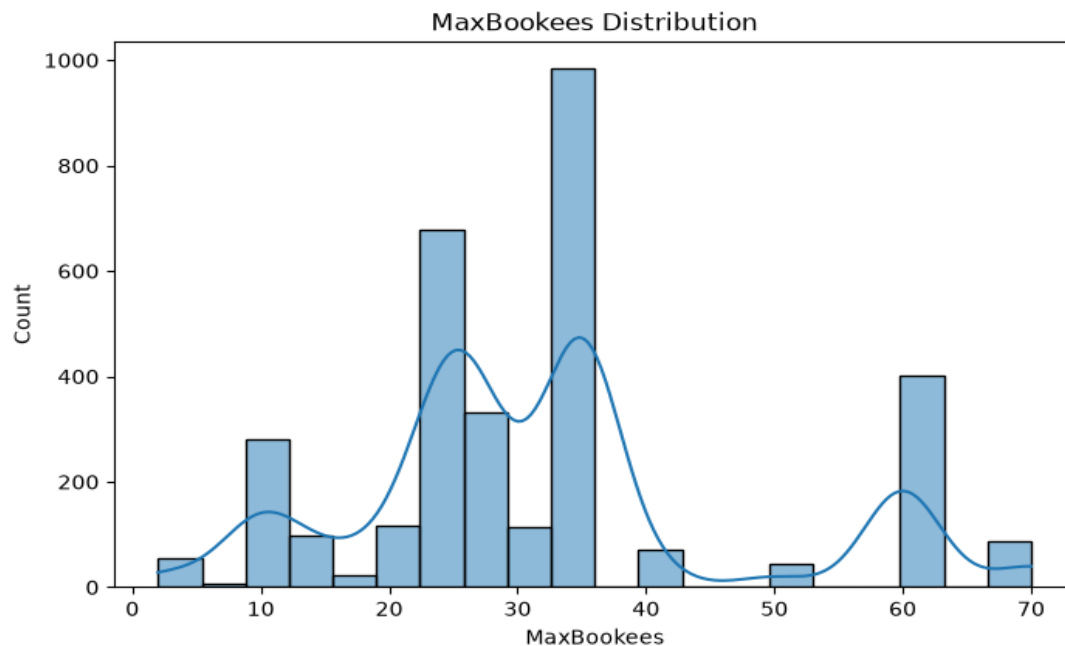
Observation

Most fitness classes receive bookings within a normal range. However, a few classes have significantly higher bookings than the rest, indicating the presence of outliers.

Business Insight

Classes with exceptionally high bookings indicate strong customer demand. These classes can be considered for premium pricing, additional sessions, or increased class capacity to maximize revenue.

3. MaxBookees (Histogram)



Objective

To analyze the distribution of class capacity.

Observation

Most fitness classes have a maximum capacity between 20 and 35 participants. Only a few classes have larger capacities of around 60 to 70 participants, while very few classes have low capacities.

Business Insight

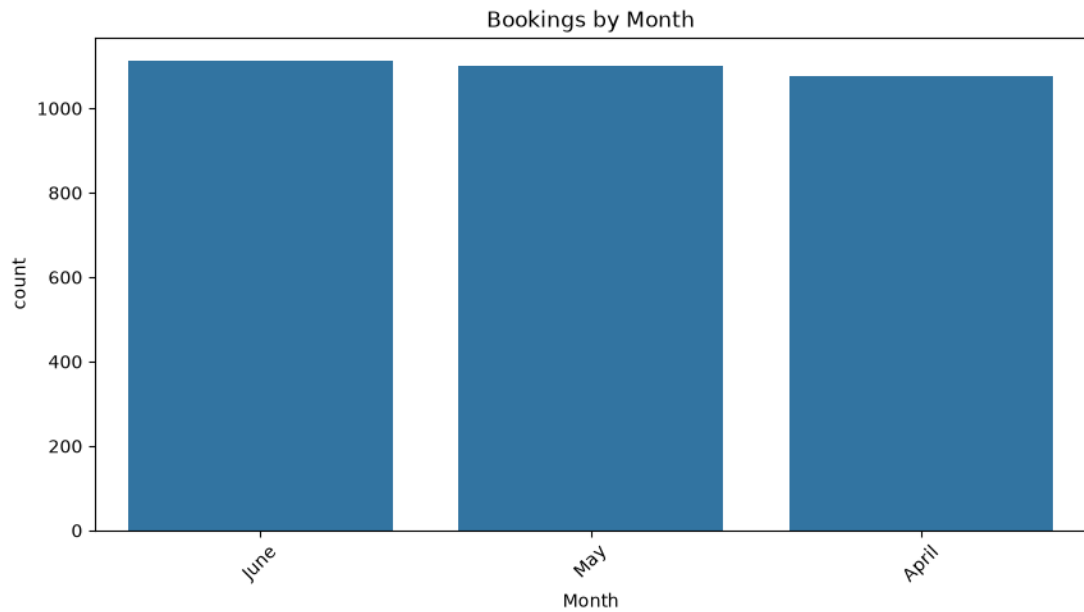
Most classes are planned for medium-sized groups, which helps maintain efficient operations. High-capacity classes should focus on achieving better occupancy to maximize revenue and improve resource utilization.

Time-Based Analysis

Purpose

Time-based analysis helps understand how customer bookings change across different months, days, hours, and time slots. These insights are useful for identifying peak demand periods and supporting dynamic pricing decisions.

1. Bookings by Month



Objective

To analyze customer booking trends across different months.

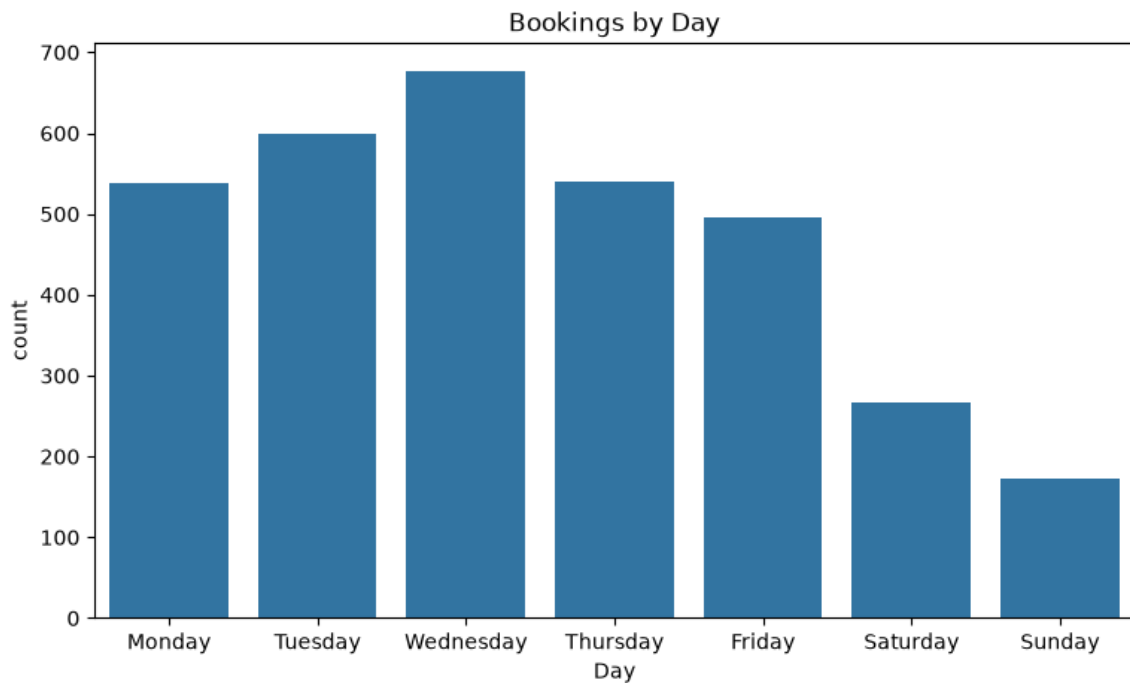
Observation

Bookings are distributed almost evenly across April, May, and June. June records the highest number of bookings, while April has slightly fewer bookings compared to the other two months.

Business Insight

Customer demand remains stable across all three months. Since June shows slightly higher demand, it may be suitable for premium pricing, while the overall consistent demand suggests a stable customer base.

2. Bookings by Day



Objective

To analyze booking patterns across different days of the week.

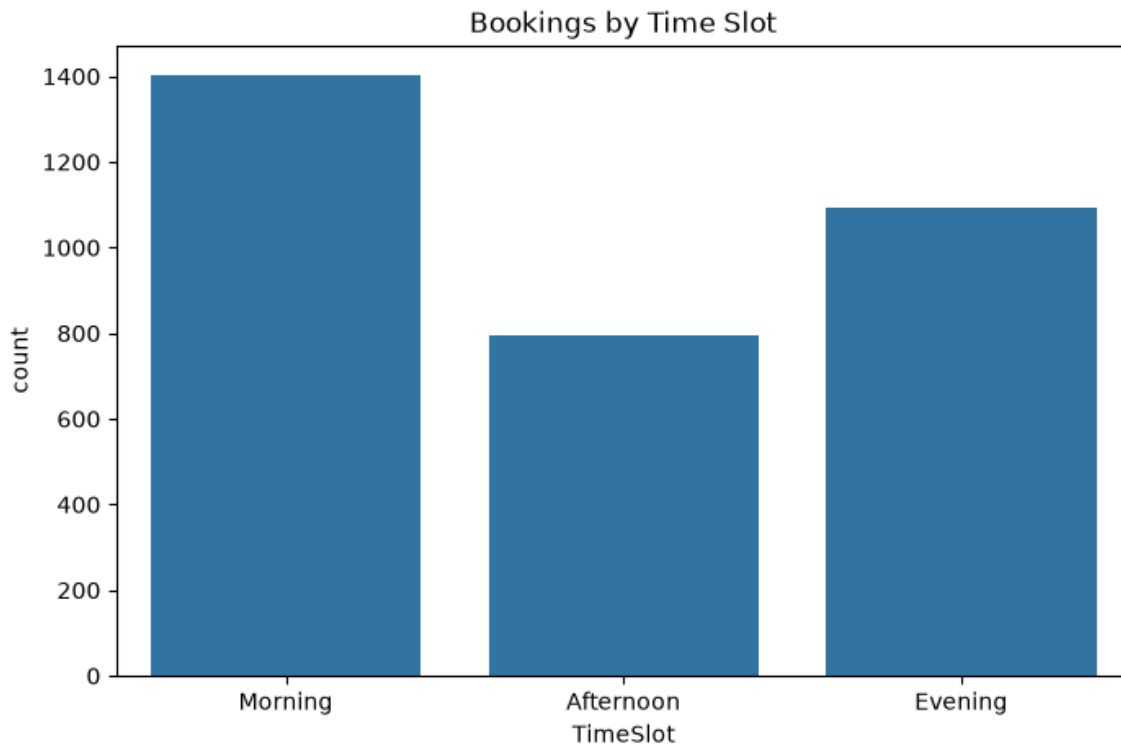
Observation

Wednesday records the highest number of bookings, followed by Tuesday. Saturday and Sunday have the lowest bookings, with Sunday showing the least customer demand.

Business Insight

Weekdays experience higher customer demand compared to weekends. Premium pricing can be considered for high-demand weekdays, while promotional offers or special weekend packages may help increase bookings on Saturdays and Sundays.

3. Bookings by Time Slot



Objective

To compare customer bookings across Morning, Afternoon, and Evening time slots.

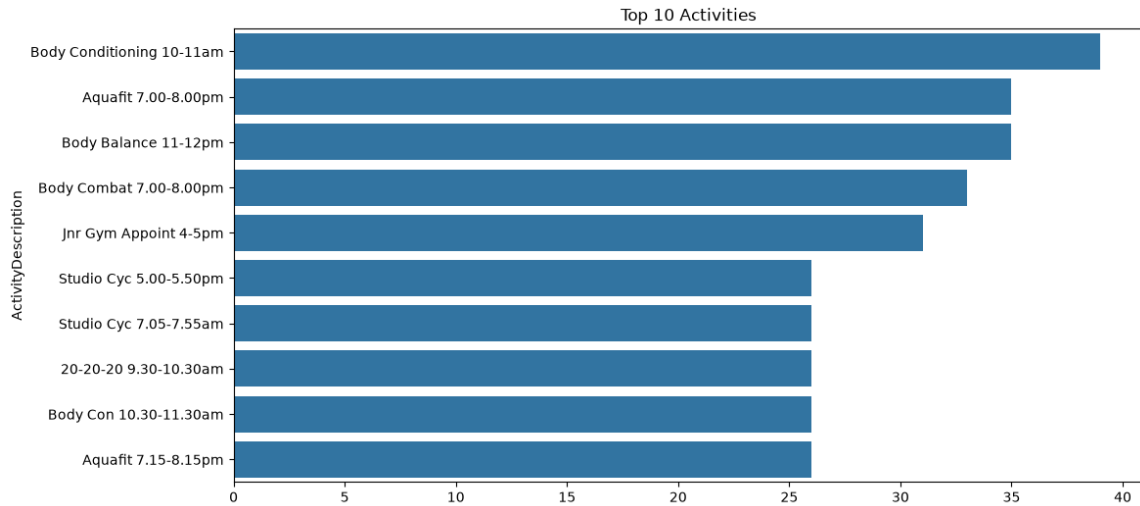
Observation

Morning time slot has the highest number of bookings, followed by Evening. Afternoon records the lowest number of bookings among all three time slots.

Business Insight

Morning and Evening time slots have higher customer demand and are suitable for premium pricing. Afternoon classes may require promotional offers or discounts to improve bookings and occupancy.

4. Top 10 Activities



Objective

To identify the fitness activities with the highest number of bookings.

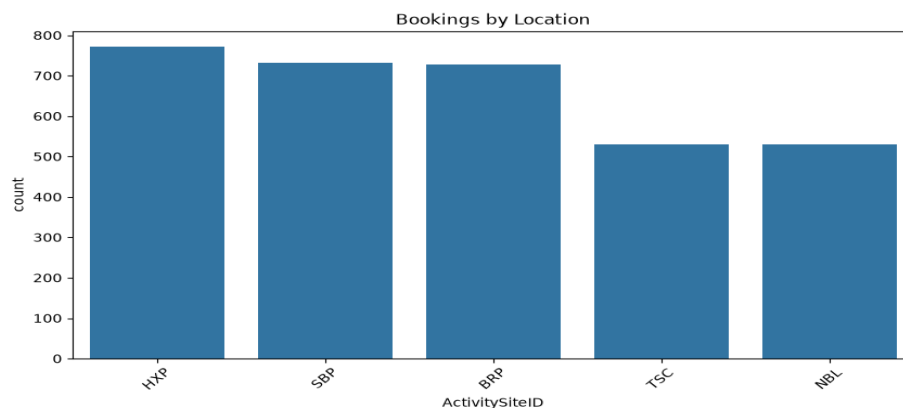
Observation

Body Conditioning (10–11 AM) has the highest number of bookings among all activities. AquaFit, Body Balance, and Body Combat classes also receive a high number of customer bookings, showing strong demand.

Business Insight

The most popular activities have consistent customer demand and are suitable for premium pricing or additional class sessions. Activities with lower bookings may require promotional offers or schedule optimization to improve participation.

5. Bookings by Location



Objective

To compare customer bookings across different fitness center locations.

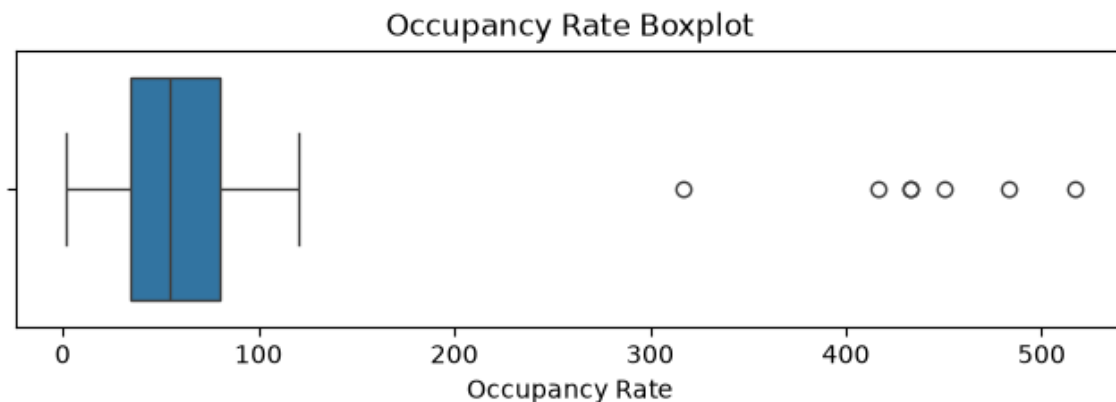
Observation

HXP records the highest number of bookings among all locations. SBP and BRP also show strong customer demand, while TSC and NBL have comparatively fewer bookings.

Business Insight

High-performing locations such as HXP can support premium pricing due to strong demand. Locations with lower bookings, such as TSC and NBL, may benefit from promotional offers, local marketing campaigns, or schedule optimization to improve customer participation.

6. Occupancy Rate (Boxplot)



Objective

To analyze the distribution of occupancy rates and identify any unusual values across fitness classes.

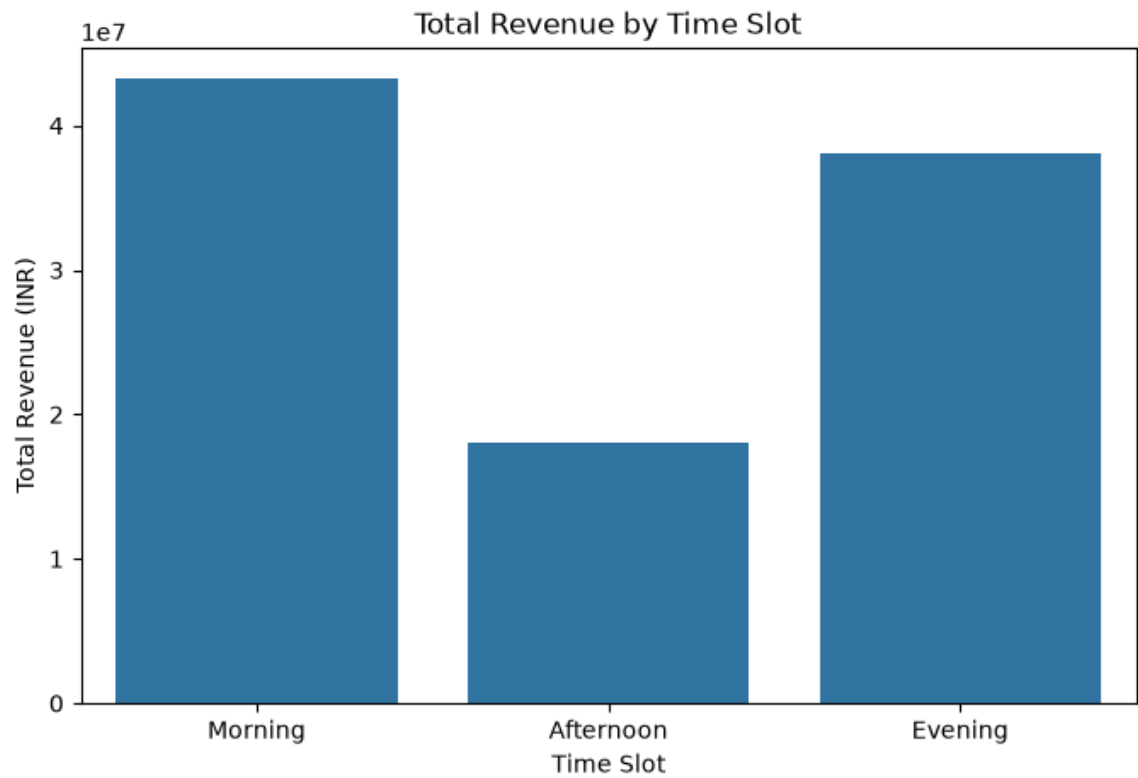
Observation

Most fitness classes have occupancy rates within a normal range. However, a few classes show extremely high occupancy rates, indicating the presence of outliers.

Business Insight

Classes with very high occupancy indicate strong customer demand and may benefit from premium pricing or additional sessions. The unusually high occupancy values should also be reviewed to ensure the booking data is accurate.

7._ Revenue by Time Slot



Objective

To compare the total revenue generated across different time slots.

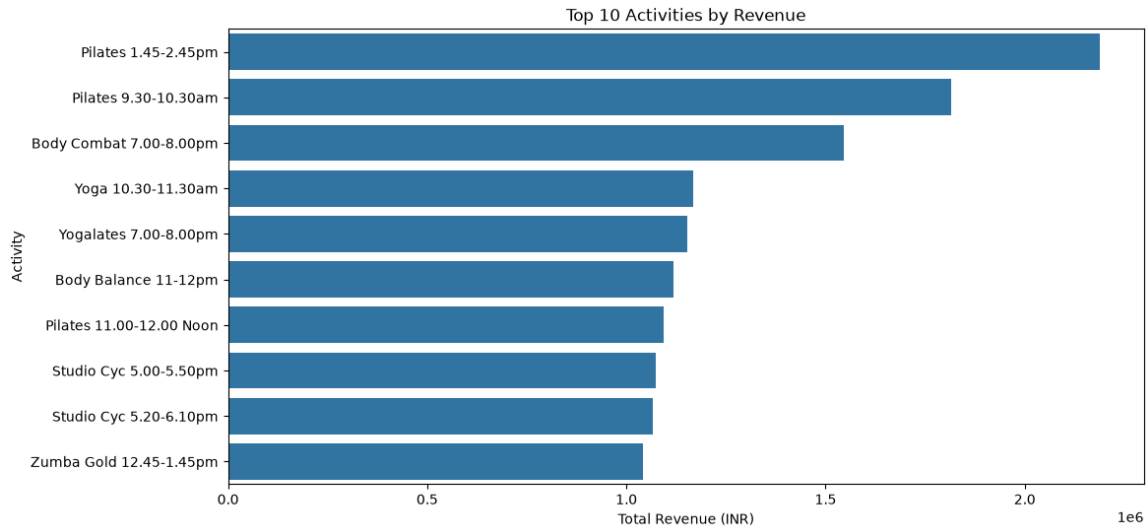
Observation

Morning time slot generates the highest revenue, followed by Evening. Afternoon contributes the lowest revenue among all time slots.

Business Insight

Morning and Evening time slots have higher revenue potential and are suitable for premium pricing. Afternoon time slots may require promotional offers or discounts to increase bookings and improve overall revenue.

8. Top 10 Activities by Revenue



Objective

To identify the fitness activities that generate the highest total revenue.

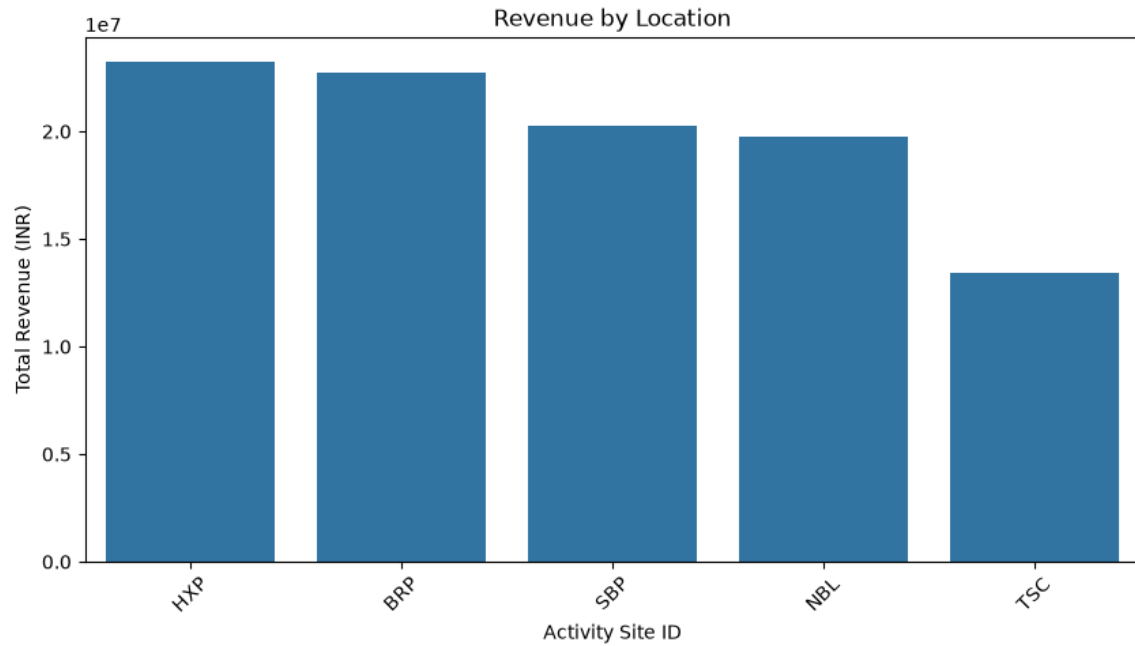
Observation

Pilates (1:45–2:45 PM) generates the highest revenue, followed by Pilates (9:30–10:30 AM) and Body Combat (7:00–8:00 PM). These activities contribute significantly more revenue than the other fitness classes.

Business Insight

High-revenue activities should be prioritized through premium pricing, additional class sessions, and effective resource allocation. Activities generating comparatively lower revenue can be supported with promotional offers or schedule optimization to improve their performance.

9. Revenue by Location



Objective

To compare the total revenue generated across different fitness center locations.

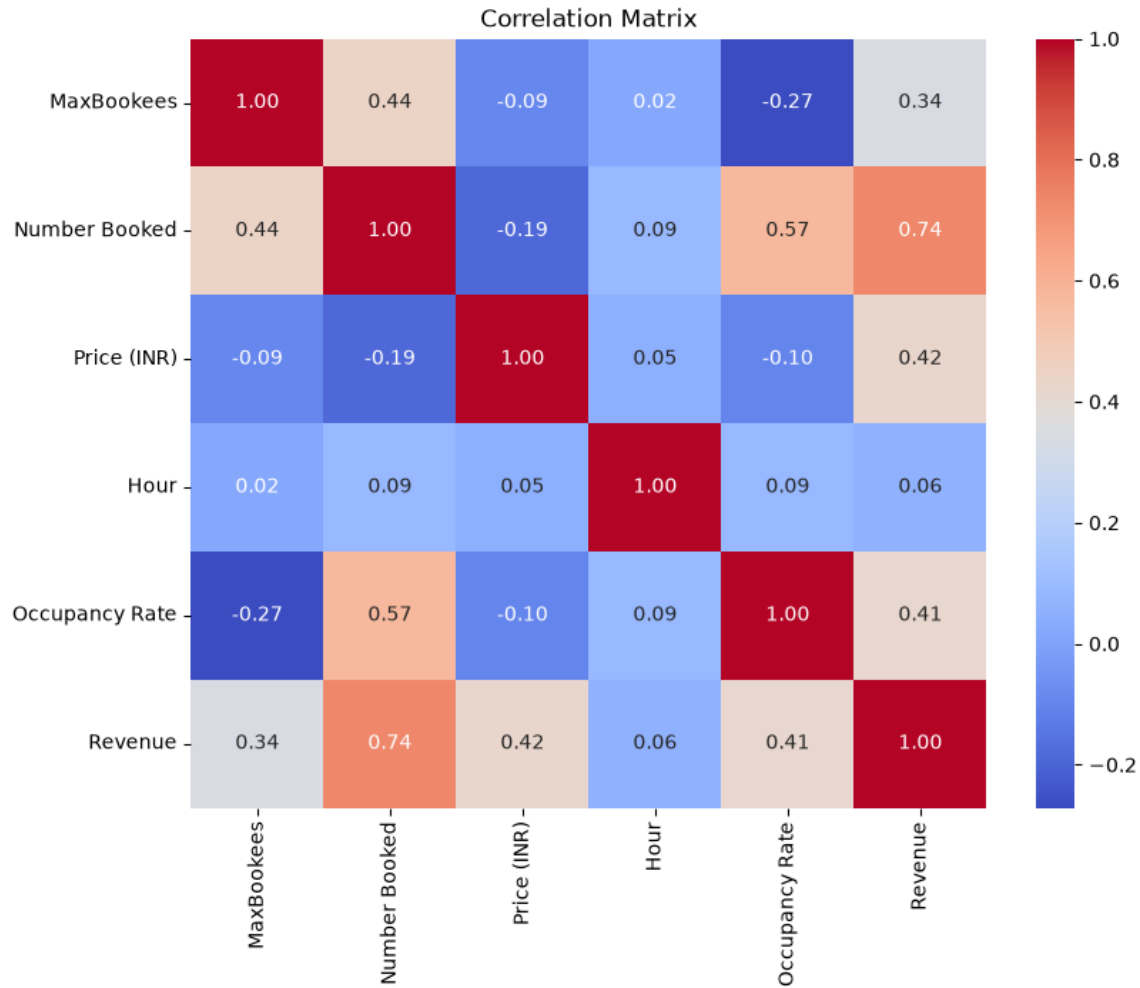
Observation

HXP generates the highest total revenue, followed by BRP and SBP. NBL records moderate revenue, while TSC generates the lowest revenue among all locations.

Business Insight

High-performing locations such as HXP and BRP have strong revenue potential and can support premium pricing strategies. Locations with lower revenue, such as TSC, may require targeted marketing, promotional offers, or schedule optimization to improve customer demand and overall revenue.

10. Correlation Analysis



Objective

To understand the relationship between key numerical features such as Price, Number Booked, MaxBookees, Occupancy Rate, Hour, and Revenue. This analysis helps identify the factors that influence customer demand and revenue.

Observation

- Revenue has the strongest positive correlation with Number Booked (0.74), indicating that more bookings lead to higher revenue.
- Occupancy Rate shows a moderate positive correlation with Number Booked (0.57).
- Price has a moderate positive correlation with Revenue (0.42), suggesting that pricing also contributes to revenue generation.

-Hour has a very weak correlation with the other variables, indicating that class timing alone has little impact.

-MaxBookees has a moderate positive relationship with Number Booked (0.44).

Business Insight

The correlation analysis shows that customer bookings are the primary driver of revenue. Improving occupancy and increasing bookings will have a greater impact on revenue than changing prices alone. These findings support the use of demand-based dynamic pricing, where prices can be adjusted based on booking trends, occupancy levels, and customer demand.

Key Business Insights

1. Morning time slot recorded the highest bookings and revenue, indicating that customers prefer attending fitness classes in the morning.
2. Wednesday showed the highest number of bookings, while Sunday had the lowest demand, suggesting that customer participation is stronger on weekdays than weekends.
3. Most fitness class prices are between ₹1,000 and ₹3,000, showing that the organization mainly follows a standard pricing structure.
4. Body Conditioning (10–11 AM) was the most booked activity, while Pilates sessions generated the highest revenue, showing that the most popular class is not always the highest revenue-generating class.
5. HXP generated the highest bookings and total revenue, making it the best-performing location among all fitness centers.
6. Revenue has the strongest positive relationship with the number of bookings (Correlation = 0.74). This indicates that increasing bookings has the greatest impact on revenue growth.
7. Occupancy Rate is positively related to Number Booked (Correlation = 0.57), showing that classes with more bookings generally achieve better seat utilization.
8. A few classes showed exceptionally high bookings and occupancy rates, indicating high-demand sessions that may require additional class schedules or capacity planning.

Recommendations

9. Apply premium pricing for Morning and Evening time slots, as these periods consistently generate higher bookings and revenue.
10. Introduce promotional offers for Afternoon classes and weekends, especially Sunday, to improve customer participation during low-demand periods.
11. Increase the availability of high-demand activities, such as Body Conditioning and Pilates, by adding more sessions where operationally feasible.
12. Use successful practices from the HXP location to improve performance at lower-performing locations such as TSC through better scheduling or targeted marketing.
13. Monitor high-occupancy classes regularly and consider increasing class capacity or adding extra batches if demand continues to exceed available seats.
14. Build the dynamic pricing model primarily around booking demand, occupancy rate, and time slot, since these factors showed the strongest relationship with revenue in the EDA.

Conclusion

This Exploratory Data Analysis provides valuable insights into customer demand, booking behavior, pricing patterns, occupancy, revenue, and location performance. These findings will support the development of a demand-based dynamic pricing model for fitness classes.

End of Report